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Question 1. Euclidean, My Dear Watson

(a)

- Equation already in standard form.
- We start by finding $\gcd(38, 7)$ (tableau on the right):

$$\begin{aligned}
 \gcd(38, 7) &= \gcd(7, 38 \bmod 7) = \gcd(7, 3) \mid 38 = 5 \cdot 7 + 3 \\
 &= \gcd(3, 7 \bmod 3) = \gcd(3, 1) \mid 7 = 2 \cdot 3 + 1 \\
 &= \gcd(1, 3 \bmod 1) = \gcd(1, 0) \mid
 \end{aligned}$$

Next, solve the equations for r :

$$\begin{aligned}
 3 &= 38 - 5 \cdot 7 \\
 1 &= 7 - 2 \cdot 3
 \end{aligned}$$

Then, we substitute backwards:

$$\begin{aligned}
 1 &= 7 - 2 \cdot 3 \\
 &= 7 - 2 \cdot (38 - 5 \cdot 7) \\
 &= -2 \cdot 38 + 11 \cdot 7
 \end{aligned}$$

Thus 11 is the multiplicative inverse of 7 modulo 38.

Finally multiplying both sides by 11,

$$x \equiv 11 \cdot 7x \equiv 11 \cdot 5 \equiv 17 \pmod{38}$$

- $x = 17 + 38k$ for any integer k .
- The original modular equation was already in standard form.
- If we let $k = 1000$, then we get a solution $x = 17 + 38 \cdot 1000 = 3817$, which is ≥ 1000 .

(b)

1.

$$62x - 6 \equiv 4 - 25x \pmod{50}$$

Add 6 to both sides:

$$62x \equiv 10 - 25x \pmod{50}$$

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Add $25x$ to both sides:

$$87x \equiv_{50} 10$$

2. We start by finding $\gcd(87, 50)$ (tableau on the right):

$$\begin{aligned} \gcd(87, 50) &= \gcd(50, 87 \bmod 50) = \gcd(50, 37) \mid 87 = 1 \cdot 50 + 37 \\ &= \gcd(37, 50 \bmod 37) = \gcd(37, 13) \mid 50 = 1 \cdot 37 + 13 \\ &= \gcd(13, 37 \bmod 13) = \gcd(13, 11) \mid 37 = 2 \cdot 13 + 11 \\ &= \gcd(11, 13 \bmod 11) = \gcd(11, 2) \mid 13 = 1 \cdot 11 + 2 \\ &= \gcd(2, 11 \bmod 2) = \gcd(2, 1) \mid 11 = 5 \cdot 2 + 1 \\ &= \gcd(1, 2 \bmod 1) = \gcd(1, 0) \mid \end{aligned}$$

Next, we solve the equations for r :

$$\begin{aligned} 37 &= 87 - 1 \cdot 50 \\ 13 &= 50 - 1 \cdot 37 \\ 11 &= 37 - 2 \cdot 13 \\ 2 &= 13 - 1 \cdot 11 \\ 1 &= 11 - 5 \cdot 2 \end{aligned}$$

Then, we substitute backwards:

$$\begin{aligned} 1 &= 11 - 5 \cdot 2 \\ &= 11 - 5 \cdot (13 - 1 \cdot 11) \\ &= -5 \cdot 13 + 6 \cdot 11 \\ &= -5 \cdot 13 + 6 \cdot (37 - 2 \cdot 13) \\ &= 6 \cdot 37 - 17 \cdot 13 \\ &= 6 \cdot 37 - 17 \cdot (50 - 1 \cdot 37) \\ &= -17 \cdot 50 + 23 \cdot 37 \\ &= -17 \cdot 50 + 23 \cdot (87 - 1 \cdot 50) \\ &= 23 \cdot 87 - 40 \cdot 50 \end{aligned}$$

Thus 23 is the multiplicative inverse of 87 modulo 50.

Finally, multiplying both sides by 23,

$$x \equiv_{50} 23 \cdot 87x \equiv_{50} 23 \cdot 10 \equiv_{50} 30$$

3. $x = 30 + 50k$ for any integer k .

4. All operations applied were reversible, so the solutions are the same.

5. If we let $k = 1000$, then we get a solution $x = 30 + 50 \cdot 1000 = 50030$, which is ≥ 1000 .

Question 2. Sum Kind of Wonderful**(a)**

Let $P(n)$ be " $\sum_{i=0}^n (5 \cdot 6^i + 3) = 6^{n+1} + 3n + 2$ ". We will show $P(n)$ is true for all non-negative integers by induction.

Base Case: $\sum_{i=0}^0 (5 \cdot 6^i + 3) = 5 \cdot 6^0 + 3 = 8 = 6^{0+1} + 3(0) + 2$, so $P(0)$ is true.

Suppose that $P(k)$ is true for some arbitrary integer $k \geq 0$. We can see that

$$\begin{aligned}\sum_{i=0}^{k+1} (5 \cdot 6^i + 3) &= \sum_{i=0}^k (5 \cdot 6^i + 3) + (5 \cdot 6^{k+1} + 3) \\ &= 6^{k+1} + 3k + 2 + 5 \cdot 6^{k+1} + 3 \quad \text{by IH} \\ &= 6 \cdot 6^{k+1} + 3k + 5 \\ &= 6^{(k+1)+1} + 3(k+1) + 2\end{aligned}$$

Therefore, $P(k+1)$ is true.

Thus, $P(n)$ is true for all non-negative integers by induction.

(b)

Let $P(n)$ be " $3^n \geq 2n + 1$ ". We will show $P(n)$ is true for all integers $n \geq 1$ by induction.

Base Case: $3^1 = 3 \geq 2(1) + 1$, so $P(1)$ is true.

Suppose that $P(k)$ is true for some arbitrary integer $k \geq 1$. We can see that

$$\begin{aligned}3^{k+1} &= 3^k \cdot 3 \\ &\geq (2k + 1) \cdot 3 \quad \text{by IH} \\ &= 6k + 3 \\ &\geq 2k + 2 + 1 \\ &= 2(k + 1) + 1\end{aligned}$$

Therefore, $P(k+1)$ is true.

Thus, $P(n)$ is true for all integers $n \geq 1$ by induction.

Question 3. Barking Up the Strong Tree

(a)

Let $P(n)$ be " $f(n) = b^n$ ". We will show $P(n)$ is true for all non-negative integers by strong induction.

Base Cases:

- $f(0) = 1 = b^0$, so $P(0)$ is true.
- $f(1) = b = b^1$, so $P(1)$ is true.

Suppose that $P(0) \wedge \dots \wedge P(k)$ is true for some arbitrary integer $k \geq 0$. We can see that

$$\begin{aligned} f(k+1) &= (b-1) \cdot f((k+1)-1) + b \cdot f((k+1)-2) && \text{def of } f \\ &= (b-1) \cdot b^k + b \cdot f(k-1) && \text{by IH} \\ &= (b-1) \cdot b^k + b \cdot b^{k-1} && \text{by IH} \\ &= b \cdot b^k - b^k + b^k \\ &= b \cdot b^k \\ &= b^{k+1} \end{aligned}$$

Therefore, $P(k+1)$ is true.

Thus, $P(n)$ is true for all non-negative integers by strong induction.

Question 4. Optional Practice Problems

(a)

(Did work on scratch paper)

i. 15 is the multiplicative inverse of 15 modulo 28.

Multiplying both sides by 15,

$$x \equiv_{28} 15 \cdot 15x \equiv_{28} 15 \cdot 14 \equiv_{28} 14$$

ii. Standard form: $12x \equiv_7 3$

3 is the multiplicative inverse of 12 modulo 7.

Multiplying both sides by 3,

$$x \equiv_7 3 \cdot 12x \equiv_7 3 \cdot 3 \equiv_7 2$$

(b)

i. Let a and b be arbitrary integers, and m and n be arbitrary positive integers.

Suppose $\gcd(m, n) = 1$.

By applying Bezout's Theorem, we get $\gcd(m, n) = sm + tn$, for some integers s and t . Then, substituting $\gcd(m, n)$, we get $sm + tn = 1$.

Notice that we can rearrange the equation to $sm - 1 = -tn$. Since there is an integer $q = -t$ such that $sm - 1 = qn$, by the definition of divides, $n \mid (sm - 1)$. Then, by the definition of modulo, $sm \equiv_n 1$. If we multiply both sides by b , we get $bsm \equiv_n b$.

Similarly, we can rearrange the equation to $tn - 1 = -sm$. Since there is an integer $q = -s$ such that $tn - 1 = qm$, by the definition of divides, $m \mid (tn - 1)$. Then, by the definition of modulo, $tn \equiv_m 1$. If we multiply both sides by a , we get $atn \equiv_m a$.

Now let $x = bsm + atn$. Under modulo m , the bsm term becomes 0, so $x \equiv_m atn \equiv_m a$.

Similarly, under modulo n , the atn term becomes 0, so $x \equiv_n bsm \equiv_n b$.

Since we have found an integer x that satisfies both $x \equiv_m a$ and $x \equiv_n b$, and a , b , m , and n were arbitrary, we have proven the claim.

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(c)

i. Let $P(n)$ be " $6 \mid (7^n - 1)$ ". We will show $P(n)$ is true for all integers $n \geq 1$ by induction.

Base Case: By the definition of divides, for $P(1)$ to be true, there must be an integer q such that $7^1 - 1 = q \cdot 6$. $7^1 - 1 = 6 = 1 \cdot 6$, so $P(1)$ is true.

Suppose that $P(k)$ is true for some arbitrary integer $k \geq 1$. We see that

$$\begin{aligned}
 6 \mid (7^k - 1) & \quad \text{by IH} \\
 7^k - 1 &= q \cdot 6 & \text{def of divides} \\
 7 \cdot (7^k - 1) &= 7 \cdot q \cdot 6 & \text{multiply by 7} \\
 7 \cdot 7^k - 7 &= 7 \cdot q \cdot 6 \\
 7^{k+1} &= 7 + 7 \cdot q \cdot 6 \\
 7^{k+1} - 1 &= 7 - 1 + 7 \cdot q \cdot 6 & \text{subtract 1} \\
 7^{k+1} - 1 &= 6 + 7 \cdot q \cdot 6 \\
 7^{k+1} - 1 &= (7q + 1) \cdot 6 \\
 6 \mid (7^{k+1} - 1) & \quad \text{def of divides}
 \end{aligned}$$

Therefore, $P(k + 1)$ is true.

Thus, $P(n)$ is true for all positive integers ($n \geq 1$) by induction.

ii. Let $P(n)$ be " $\text{Even}(n) \vee \text{Odd}(n)$ ". We will show $P(n)$ is true for all positive integers by induction.

Base Case:

- By the definition of Even, an integer n is even if $n = 2k$ for some integer k . Since there is no integer k such that $1 = 2k$, $\text{Even}(1)$ is false.
- By the definition of Odd, an integer n is odd if $n = 2k + 1$ for some integer k . Since $1 = 2(0) + 1$, $\text{Odd}(1)$ is true.
- Therefore, $\text{Even}(1) \vee \text{Odd}(1)$ is true, so $P(1)$ is true.

Suppose that $P(k)$ is true for some arbitrary integer $k \geq 1$. Then k is Even or Odd by the definition of P . We proceed by cases...

If k is Even, then by the definition of Even, there is some integer s for which $k = 2s$. Then $k + 1 = 2s + 1$, so by the definition of Odd, $k + 1$ is Odd. Then the statement $\text{Even}(k + 1) \vee \text{Odd}(k + 1)$ is also true.

If k is Odd, then by the definition of Odd, there is some integer s for which $k = 2s + 1$. Then $k + 1 = 2s + 2 = 2(s + 1)$, so by the definition of Even, $k + 1$ is Even. Then the statement $\text{Even}(k + 1) \vee \text{Odd}(k + 1)$ is also true.

Since our cases were exhaustive, we have shown that $\text{Even}(k + 1) \vee \text{Odd}(k + 1)$ is true.

Therefore, $P(k + 1)$ is true.

Thus, $P(n)$ is true for all positive integers ($n \geq 1$) by induction.